1.	**Name of Course :**	High Voltage Engineering				
	Course Code :	EEE 304				
2.	**Synopsis :**	This subject is intended to give the students deep knowledge about high voltage engineering. It focuses on the phenomena of high voltage surges and insulation coordination for power systems, characteristics of conduction and breakdown of gas, liquid and solid dielectrics. Generation of high voltages, their measurement and testing technique for high voltage components. In this subject, the student are also exposed to lightning phenomena and their protection				
3.	**Name(s) of academic staff :**	Mohd Faris bin Abdullah				
4.	**Semester and Year offered :**	**Semester**	1	**Year**	3	
5.	**Credit Value :**	3				
6.	**Prerequisite/ co-requisite: (if any)**	Electrical Engineering				

7.	**Course Learning Outcomes (CLO) : At the end of the course the students will be able to: (example) - explain the basic principles of immunisation (C2,PLO1)**	**Bloom's Taxonomy Level**	**PLO**
	CLO1 Analyze the methods of generations and measurements of high voltage AC, DC and impulse generator	C4	2
	CLO2 Illustrate the various breakdown mechanism and applications of vacuum, liquid, solid and composite dielectrics	C4	2
	CLO3 Determine the over-voltage phenomena, insulation coordination in power system, types of high voltage testing for electrical apparatus and non-destructive testing of materials	C5	4
	CLO4		
	CLO5		

8. Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment :
Please select the Learning Outcome Domain (LOD) for each PLO in the cells above it. E.g. PLO1 - Knowledge, PLO2 - Cognitive, PLO3 - Practical Skills

Course Learning Outcomes (CLO)	**Programme Learning Outcomes (PLO)**												**Learning and Teaching Method**	**Assessment Method**
	Engineering Knowledge	Problem Analysis	Design/ Development of Solutions	Investigation	Modern Tool Usage	The Engineer and Society	Environmental and Sustainability	Ethics	Individual and Team Work	Communication	Project Management and Finance	Life Long Learning		
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**		
CLO1		√											Lecture, Tutorial, Practical	Laboratory report, Final Examination
CLO2		√											Lecture, Tutorial	Midterm Examination, Final Examination
CLO3				√									Lecture, Tutorial	Assignment, Final Examination
CLO4														
CLO5														

Indicate the relevancy between the CLO and PLO by ticking "√" the appropriate relevant box.

(This description must be read together with Standards 2.1.2 , 2.2.1 and 2.2.2 in Area 2 - pages 16 & 18)

9.	**Transferable Skills (if applicable) (Skills learned in the course of study which can be useful and utilized in other settings)**	**1**	Cognitive skills
		2	

		3	
		4	
		5	

10. Distribution of Student Learning Time (SLT)

Course Content Outline	CLO*	Teaching and Learning Activities						SLT
		Guided Learning (F2F)				Guided Learning (NF2F) eg: e-Learning	Independent Learning (NF2F)	
		L	T	P	O			
1. Introduction to high voltage engineering - Electric Field Stresses - Insulator Materials - Characteristics Of Gases, Vacuum, Liquid And Solids Dielectrics - Estimation & Control Of Electric Stress, Surge Voltages And Their Distribution & Control	2	2	1				3	**6**
2. Conduction and breakdown in solid dielectric, gases and liquid - Conduction & breakdown in gases - Conduction & breakdown in liquid dielectrics - Conduction and Breakdown in solid dielectricsc	2	6	3				9	**18**
3, Insulation coordination and application of insulating materials - Application of Insulating Materials In Power Transformer - Application of Insulating Materials In Rotating Machines - Application of Insulating Materials In Circuit Breaker - Application of Insulating Materials In Power Capacitors - Application of Insulating Materials In High Voltage bushing - Application of Insulating Materials In Small Machines	3	4	2				6	**12**
4. Generation of high voltages and currents - Generation of High DC Voltages - Generation of High AC Voltages - Generation of Impulse Voltages - Generation of Impulse Currents	1	4	1	4			9	**18**
5. Over-voltage phenomenon - Over-voltage phenomenon & insulation coordination in electric power systems	3	4	2				6	**12**
6. Non-destructive insulation testing - Non-destructive testing of materials & electrical apparatus - High voltage testing of electrical apparatus	1	4	2				6	**12**
7. Lightning phenomena and their protection - Lighning introudction - Lightning protection system and coordination	3	4	2				6	**12**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
	Total	**28**	**13**	**4**	**0**	**0**	**45**	**90**

	Continuous Assessment		Percentage (%)	F2F	NF2F	SLT
1	Assignment		20		4	4
2	Laboratory Report		10	2	4	6
3	Midterm Examination		20	2	6	8
4						0
5						0
6						0
7						0
		Total	**50**	**4**	**14**	**18**

	Final Assessment		Percentage (%)	F2F	NF2F	SLT
1	Final Examination		50	2	10	12
2						0
3						0
4						0
5						0
		Total	**50**	**2**	**10**	**12**

****Please tick (√) if this course is Latihan Industri/ Clinical Placement/ Practicum/ WBL using Effective Learning Time(ELT) of 50%**

	GRAND TOTAL SLT	**120**

L = Lecture, T = Tutorial, P= Practical, O= Others, F2F=Face to Face, NF2F=Non Face to Face

**Indicate the CLO based on the CLO's numbering in Item 8.*

11	**Identify special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room, etc)**	Multisim, Pspice, Psim, Matlab/Simulink, EMTP-ATP and PSCAD

12	**References (include required and further readings, and should be the most current)**	**Required References** • E. Kuffel, W.S. Zaengl, J. Kuffel. (2016). 'High Voltage Engineering: Fundamentals', Newness. **Further Readings** • S. Naidu & V. Kamaraju, "High Voltage Engineering", McGrawHill, 2013. • Subir Ray. (2013). 'An Introduction to High Voltage Engineering', Prentice-Hall of India. • C.L. Wadhwa. (2010). 'High Voltage Engineering'. New Age International.
13	**Other additional information :**	